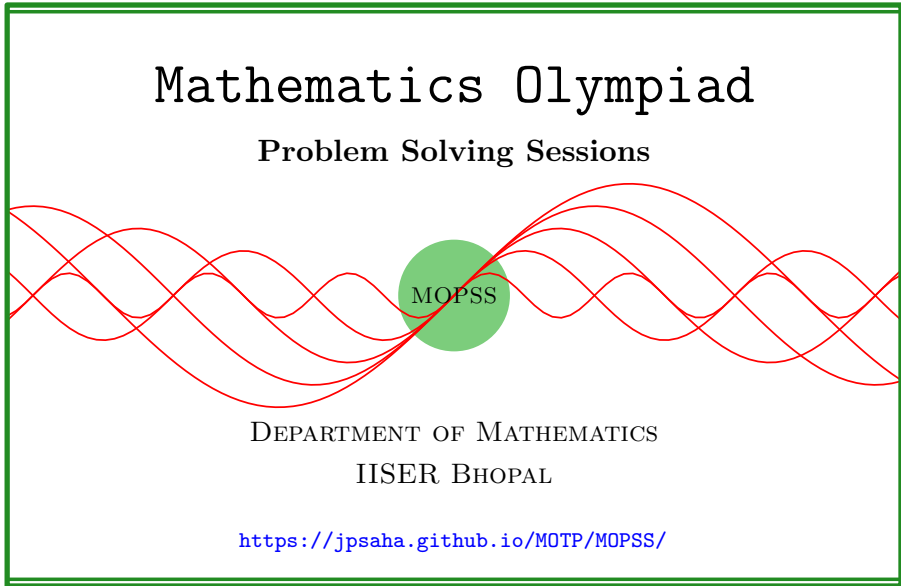


Grouping in pairs

MOPSS



Suggested readings

- Evan Chen's advice [On reading solutions](https://blog.evanchen.cc/2017/03/06/on-reading-solutions/), available at <https://blog.evanchen.cc/2017/03/06/on-reading-solutions/>.
- Evan Chen's [Advice for writing proofs/Remarks on English](https://web.evanchen.cc/handouts/english/english.pdf), available at <https://web.evanchen.cc/handouts/english/english.pdf>.
- [Notes on proofs](#) by Evan Chen from [OTIS Excerpts](#) [[Che25](#), Chapter 1].
- [Tips for writing up solutions](https://www.math.utoronto.ca/barbeau/writingup.pdf) by Edward Barbeau, available at <https://www.math.utoronto.ca/barbeau/writingup.pdf>.
- Evan Chen discusses why [math olympiads](#) are a valuable experience for [high schoolers](#) in the post on [Lessons from math olympiads](#), available at <https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/>.

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§1 Grouping in pairs

Example 1.1 (India RMO 1994 P7). Find the number of all rational numbers m/n such that

- $0 < m/n < 1$,
- m and n are relatively prime,
- $mn = 25!$.

Walkthrough —

- Determine the number of pairs of relatively prime positive integers (m, n) satisfying $mn = 25!$.
- Show that such pairs (m, n) with $m > n$, and the pairs with $m < n$ are in one-to-one correspondence.

Solution 1. Note that $25!$ factors as a product of positive integral powers of the 9 primes 2, 3, 5, 7, 11, 13, 17, 19, 23. So $25!$ has 2^9 factorizations as a product of two relatively prime positive integers, that is, there are 2^9 pairs of positive integers (m, n) such that m, n are relatively prime and $mn = 25!$. Certainly, we have either $m > n$ or $m < n$. Moreover, interchanging m, n gives a one-to-one correspondence between the factorizations $25!$ into relatively prime positive integers m, n with $m > n$ and the factorizations $25!$ into relatively prime positive integers m, n with $m < n$, that is, the map $(m, n) \mapsto (n, m)$ (that is, interchanging m and n) defines a bijection between the sets

$$\begin{aligned}
 S &= \{(m, n) | m, n \text{ are relatively prime positive integers} \\
 &\quad \text{satisfying } mn = 25! \text{ and } \frac{m}{n} < 1\}, \\
 T &= \{(m, n) | m, n \text{ are relatively prime positive integers} \\
 &\quad \text{satisfying } mn = 25! \text{ and } \frac{m}{n} > 1\}.
 \end{aligned}$$

Since $S \cup T$ has 2^9 elements, it follows that S has cardinality $2^8 = 256$. ■

Example 1.2 (Putnam 2002 A3, India INMO 2013 P4). [AF13, Exercise 4.8, p. 90] Let n be an integer greater than 1 and let T_n be the number of nonempty subsets S of $\{1, 2, \dots, n\}$ with the property that the average of the elements of S is an integer. Prove that $T_n - n$ is always even.

Walkthrough —

- (a) Show that the average of any nonempty subset of $\{1, 2, \dots, n\}$ lies between 1 and n .
- (b) Observe that it suffices to show that for any $1 \leq a \leq n$, the subsets of $\{1, 2, \dots, n\}$ with average equal to a is an odd number.
- (c) Show that for any $1 \leq a \leq n$, the nonempty subsets of $\{1, 2, \dots, n\}$ having average equal to a and not containing a , are in one-to-one correspondence with the subsets of $\{1, 2, \dots, n\}$ having average equal to a , and which contain a . (Consider the averages of $\{1, 2, 3\}$ and $\{2\}$.)

Solution 2. Let $\mathcal{T}_n$ denote the set of nonempty subsets of $\{1, 2, \dots, n\}$ having integer average. Note that the average of the elements of any subset of $\{1, 2, \dots, n\}$ lies between 1 and n . So if a nonempty subset S of $\{1, 2, \dots, n\}$ has integer average a and S does not contain a , then $S \cup \{a\}$ is also a nonempty subset of $\{1, 2, \dots, n\}$ with average a . Hence, for any integer $1 \leq a \leq n$, there is a bijection between the nonempty subsets of X with integer average a that does not contain a and the subsets of X with integer average a that contain a , but not equal to $\{a\}$. So the subsets of $\mathcal{T}_n$ with average equal to a is an odd number for any $1 \leq a \leq n$. Hence $T_n - n$ is even. ■

Example 1.3 (India RMO 2012f P6). Let S be the set $\{1, 2, \dots, 10\}$. Let A be a subset of S . We arrange the elements of A in increasing order, that is, $A = \{a_1, a_2, \dots, a_k\}$ with $a_1 < a_2 < \dots < a_k$. Define WSUM for this subset as $3(a_1 + a_3 + \dots) + 2(a_2 + a_4 + \dots)$ where the first term contains the odd numbered terms and the second the even numbered terms. (For example, if $A = \{2, 5, 7, 8\}$, WSUM is $3(2 + 7) + 2(5 + 8)$.) Find the sum of WSUMs over all the subsets of S . (Assume that WSUM for the null set is 0.)

Walkthrough —

- (a) Note that

$$\begin{aligned}\text{WSUM}(\{2, 5, 7, 8\}) &= 3(2 + 7) + 2(5 + 8), \\ \text{WSUM}(\{1, 2, 5, 7, 8\}) &= 3(1 + 5 + 8) + 2(2 + 7),\end{aligned}$$

which shows that

$$\text{WSUM}(\{2, 5, 7, 8\}) + \text{WSUM}(\{1, 2, 5, 7, 8\}) = 3 + 5(2 + 5 + 7 + 8).$$

- (b) Also note that the sum of all the elements of a subset of A of $\{2, 3, \dots, 10\}$, and the sum of all the elements of its complement in $\{2, 3, \dots, 10\}$, add up to the sum of all the elements of $\{2, 3, \dots, 10\}$.

Solution 3. Note that the subsets of $\{1, 2, \dots, 10\}$ can be obtained by considering the subsets of $\{2, 3, \dots, 10\}$, along with the union of these subsets with

$\{1\}$. Using this observation, we decompose the required sum as follows.

$$\begin{aligned}
 \sum_{A \subseteq S} \text{WSUM}(A) &= \sum_{A \subseteq \{2,3,\dots,10\}} \text{WSUM}(A) + \text{WSUM}(A \cup \{1\}) \\
 &= \sum_{A \subseteq \{2,3,\dots,10\}} \left(3 + 5 \sum_{a \in A} a \right) \\
 &= 3 \cdot 2^9 + 5 \sum_{A \subseteq \{2,3,\dots,10\}} \sum_{a \in A} a \\
 &= 3 \cdot 2^9 + \frac{5}{2} \sum_{A \subseteq \{2,3,\dots,10\}} \left(\sum_{a \in A} a + \sum_{a \in A^c} a \right) \\
 &= 3 \cdot 2^9 + \frac{5}{2} \sum_{A \subseteq \{2,3,\dots,10\}} (2 + 3 + \dots + 10) \\
 &= 3 \cdot 2^9 + \frac{5}{2} \times 54 \times 2^9 \\
 &= 138 \times 2^9.
 \end{aligned}$$

■

References

- [AF13] T. ANDREESCU and Z. FENG. *A Path to Combinatorics for Undergraduates: Counting Strategies*. Birkhäuser Boston, 2013. ISBN: 9780817681548. URL: <https://books.google.de/books?id=3mwQBwAAQBAJ> (cited p. 2)
- [Che25] EVAN CHEN. *The OTIS Excerpts*. Available at <https://web.evanchen.cc/excerpts.html>. 2025, pp. vi+289 (cited p. 1)